

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457853.7758 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.056
Transit depth (Rp/Rs)^2	3.07 +/- 1.98 [%]
Orbital Inclination (inc)	86.87 +/- 2.89
Airmass coefficient 1 (a1)	0.53 +/- 0.3
Airmass coefficient 2 (a2)	0.46 +/- 0.32
Scatter in the residuals of the lightcurve fit is	103.29 %
Best Comparison Star	None
Optimal Aperture	8.98
Optimal Annulus	19.53
Transit Duration (day)	0.0599 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.056 vs 0.1594 (deviation 0.25σ)
Duration fit vs published	0.0599 d vs 0.0512 d (deviation 1.52σ)
Mid-time O-C	-2.6 min
Depth SNR	2.8
Residual scatter	103.29 %

Light curve

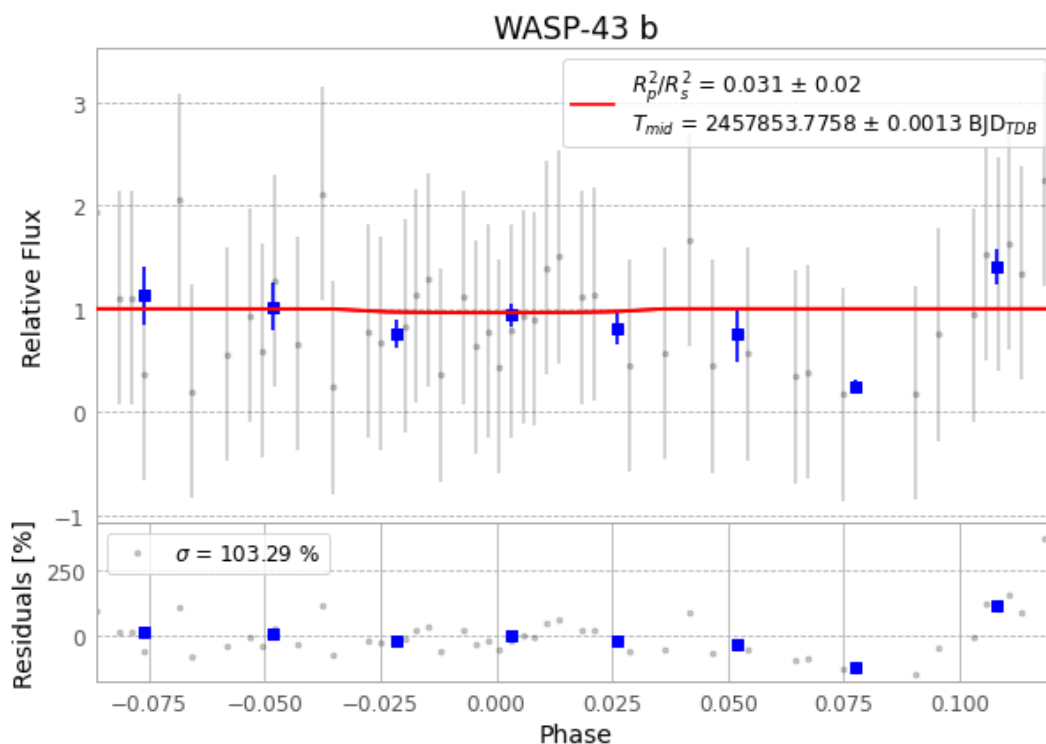


Figure 1 — FinalLightCurve_WASP-43 b_2017-04-09.png (SHA-256 497991d1f0f50a20...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [65, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c1749f705e202510...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

88 FITS frames (58 MB), WASP-43170410044812.FITS → WASP-43170410090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170410.log (6fc107e295d2...), FinalLightCurve_WASP-43 b_2017-04-09.png (497991d1f0f5...), FinalLightCurve_WASP-43 b_2017-04-09.csv (cdcd521521fc...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 18:17Z.